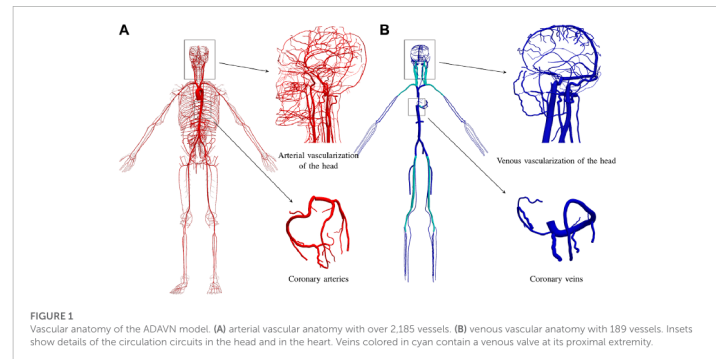


Figure 1 translation specimen - Korean

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confluence of sinuses (CoS) provides a connection among the SSS, the TS, the occipital sinus (OS) and the transverse sinus (TS). After collecting most of the blood coming from the SSS and the TS, the TS drains blood from the occipital and temporal superficial parts of the brain, through the corresponding inferior cerebral veins in that region. Then, towards the anterior part of the brain the TS connects to the sigmoid sinus (SSS) and superior petrosal sinus, which in turn connect to the inferior petrosal sinus (IPS), the cavernous and the posterior intracavernous sinuses. At that point, the basilar plexus is also connected, and provides a direct pathway to the confluence of sinuses through the marginal sinus and the OS.

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IJVs collect blood from the brain, draining from the SIS and the IPS, and from face and neck, finally arriving at the BrV.

The lateral anterior condylar vein connects the IJV with the CoS through the occipital vein, and towards the heart with the suboccipital sinus, from which the vertebral vein and the deep cervical brain branch as important collateral pathways for the blood to be drained towards the BrV.

An additional ingredient of the present model is the existence of Starling-like elements which are able to account for the venous waterfall effect between dural sinuses and cerebral veins, wherein, when a portion of the vessel collapses the flow becomes independent of the central venous pressure (Müller and Toro, 2014a). This ingredient targets the hypothesis that establishes that CBF is ruled by the difference between arterial blood pressure and intracranial pressure. Figure 2 features the (color change) interfaces at which Starling elements are placed in the ADAVN model.

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2.1.3 Arterial-venous connectivity

The criteria and hypotheses to define the connectivity between arterial and venous peripheral beds were the following:

- each terminal artery is connected to an arteriolar bed, which can connect to multiple veins;
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- the connection between arteriolar and venular beds is modeled through a purely resistive element to represent the capillary resistance.

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해부학적으로 상세한 동맥-정맥 네트워크 모델: 뇌 및 관상순환

TRANSLATED FIGURE CAPTION

ADAVN 모델의 혈관 해부 구조. (A) 2,185개가 넘는 혈관을 포함한 동맥 혈관 해부 구조. (B) 189개 혈관을 포함한 정맥 혈관 해부 구조. 삽입 그림은 머리와 심장의 순환 회로를 자세히 보여준다. 청록색으로 표시된 정맥은 근위 말단에 정맥 판막이 있다.

TRANSLATED LABELS

머리의 동맥 혈관 분포

머리의 정맥 혈관 분포

관상동맥

관상정맥

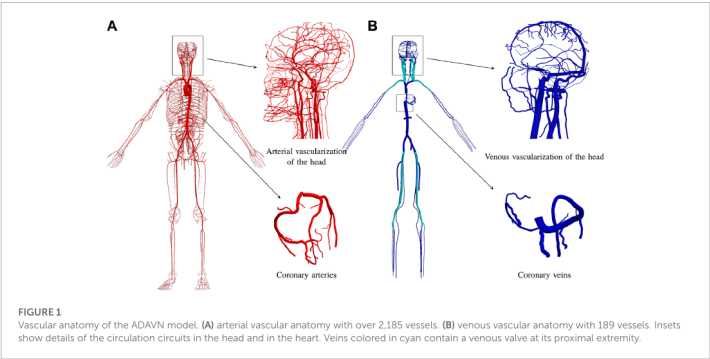
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Figure 1 translation specimen - Spanish

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Modelo anatómicamente detallado de la red arterial y venosa: circulación cerebral y coronaria

TRANSLATED FIGURE CAPTION

Anatomía vascular del modelo ADAVN. (A) Anatomía arterial con más de 2.185 vasos. (B) Anatomía venosa con 189 vasos. Los recuadros muestran detalles de los circuitos circulatorios de la cabeza y el corazón. Las venas coloreadas en cian presentan una válvula venosa en su extremo proximal.

TRANSLATED LABELS

Vascularización arterial de la cabeza

Vascularización venosa de la cabeza

Arterias coronarias

Venas coronarias

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Figure 1 translation specimen - Arabic

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نموذج مفصل تشريحيًا للشبكة الشريانية الوريدية: الدوران الدماغي والتاجي

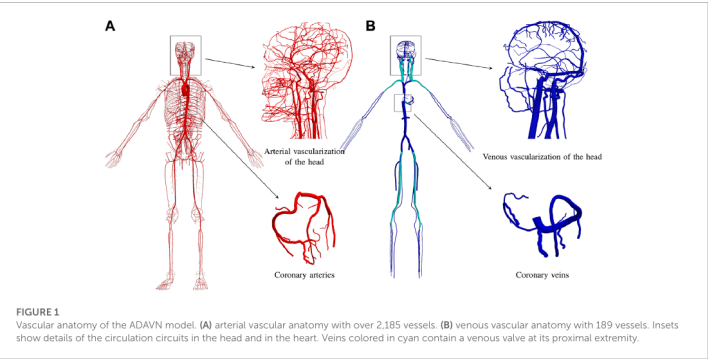
TRANSLATED FIGURE CAPTION

التشريح الوعائي لنموذج (A). التشريح الوعائي الشرياني الذي يضم أكثر من 2,185 وعاء. (B) التشريح الوعائي الوريدي الذي يضم 189 وعاء. تظهر الصور المدرجة تفاصيل الدوائر الدورانية في الرأس والقلب. تحتوي الأوردة الملونة بالأزرق السماوي على صمام وريدي عند نهايتها القريبة.

TRANSLATED LABELS

التروية الشريانية للرأس
التروية الوريدية للرأس
الشرايين التاجية
الأوردة التاجية

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